

Task 1: Data Collection & Dataset Understanding



1. Introduction

The purpose of this task is to collect or load a dataset and understand its structure before performing data cleaning, analysis, visualization or prediction. The reason I chose to perform this task is because it sets up the base for future tasks such as creating a predictive model.

For this task, I selected the Students Performance dataset published on Kaggle. This dataset is suitable for beginner data science work because it contains both categorical information and numerical information.

2. Dataset Source

The dataset used in this task is the Students Performance in Exams dataset published on Kaggle. It is publicly available and commonly used for beginner data analysis and machine learning projects. The dataset focuses on student exam performance in three subjects: math, reading, and writing.

3. Dataset Overview

The dataset contains:

Item	Value
Number of rows	1000
Number of columns	8
Missing values	0
Duplicate rows	0

Table 1: Dataset Overview Table

Each row represents one student. Each column contains either background information about the student or their exam scores.

4. Column Description

Column Name	Description	Data Type
Gender	Gender of the student	Categorical
Race/Ethnicity	Student group category	Categorical
Parental Level of Education	Highest education level of the student's parents	Categorical
Lunch	Type of lunch plan	Categorical
Test Preparation Course	Whether the student completed test preparation	Categorical
Math Score	Student's math exam score	Numerical
Reading Score	Student's reading exam score	Numerical
Writing Score	Student's writing exam score	Numerical

Table 2: Category Table

5. Missing Values and Duplicates

Upon reviewing the dataset, there were no missing values, nor duplicate rows. All 1000 students have complete information.

6. Basic Score Statistics

The dataset contains three numerical score columns: math score, reading score, and writing score.

Score Column	Average Score	Minimum Score	Maximum Score
Math Score	66.06	0	100
Reading Score	69.17	17	100
Writing Score	68.05	10	100

Table 3: Score Statistics

Based on the average scores, reading has the highest average, followed by writing and then math.

7. Task 1 Summary

In this task, I selected and reviewed the Students Performance in Exams dataset. The dataset contains 1000 rows and 8 columns. It includes student background information and exam scores in math, reading, and writing.

The dataset has no missing values nor any duplicate rows. Thus, making it suitable for the next stages of the project, including data cleaning, exploratory data analysis, visualization, and prediction.

Overall, this task helped me understand the dataset structure before continuing with the remaining data science tasks.

8. Sources

<https://www.kaggle.com/datasets/spscientist/students-performance-in-exams?resource=download>